

SPHEREx Pipeline: Source Decontamination and Spectral Extraction

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Repository:

https://github.com/andrijazupic/SPHEREx_pipeline

Mission Overview: SPHEREx

- **Mission:**
 - Spectro-Photometer for the History of the Universe, Epoch of Reionization, and Ices Explorer
- **Type:**
 - NASA Near-Infrared All-Sky Spectral Survey
- **Orbit:**
 - Sun-synchronous Earth orbit (terminator orbit)
- **Survey Method:**
 - All-sky survey repeated 4 times over 2 years

Instrumental Specifications

- **Wavelength Range:**
 - 0.75 to 5.0 μm
- **Spectral Resolution:**
 - $R \approx 41$ (for 0.75 - 2.42 μm)
 - $R \approx 135$ (for 2.42 - 5.0 μm)
- **Spatial Resolution:**
 - 6.2" per pixel
- **Observing Strategy (Spectro-Photometry):**
 - SPHEREx does not use a traditional spectrograph (like a grating or prism). Instead, it uses **Linear Variable Filters (LVFs)**
 - **The Mechanism:** As the satellite scans the sky, a target moves physically across the focal plane, sequentially passing through **~102 distinct narrow-band filter segments**.
 - **The Result:** By combining these ~102 separate photometric measurements of the same object taken at different times, we construct a **low-resolution spectrum**. This is why it is technically "spectrophotometry."

The Challenge: Large Pixel Scale

- **The Problem:**

- SPHEREx pixels (6.2") are significantly larger than typical optical/NIR survey pixels
- **Pixel Size / Resolution Comparison:**
 - Gaia: ~0.1" (effective angular pixel size)
 - Euclid: 0.1" (VIS visible) / 0.3" (NISP infrared)
 - Pan-STARRS: 0.25"
 - DESI Legacy: 0.26"
 - SDSS: 0.40"
 - Hipparcos: ~1.2" (focal plane grid period; note: used a photomultiplier, not modern CCD pixels)
 - **SPHEREx: 6.2"**

- **Consequence:**

- A single SPHEREx pixel often contains multiple **unresolved sources**

- **Requirement:**

- We must identify and **flag "contaminated" targets** where background flux can significantly bias the target spectrum

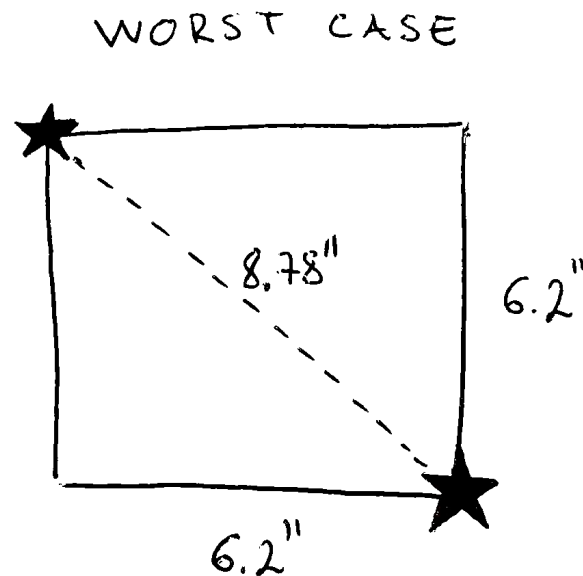
Catalog Filtering

- **Catalog-Level Cleaning:**

- We identify potential contaminants by cross-referencing our SPHEREx targets against pre-computed, high-resolution source catalogs within a **radius of 9.3"**

- **Why 9.3" radius?**

- Because the exact sub-pixel position of a target is unknown, this radius accounts for the **worst-case geometric scenario** where a target and a contaminant fall in opposite corners of the same 6.2" SPHEREx pixel (an 8.78" diagonal separation), plus an **additional ~0.5" buffer** to capture the overlapping light from their Point Spread Functions (PSF).



Multi-Survey Catalog Filtering

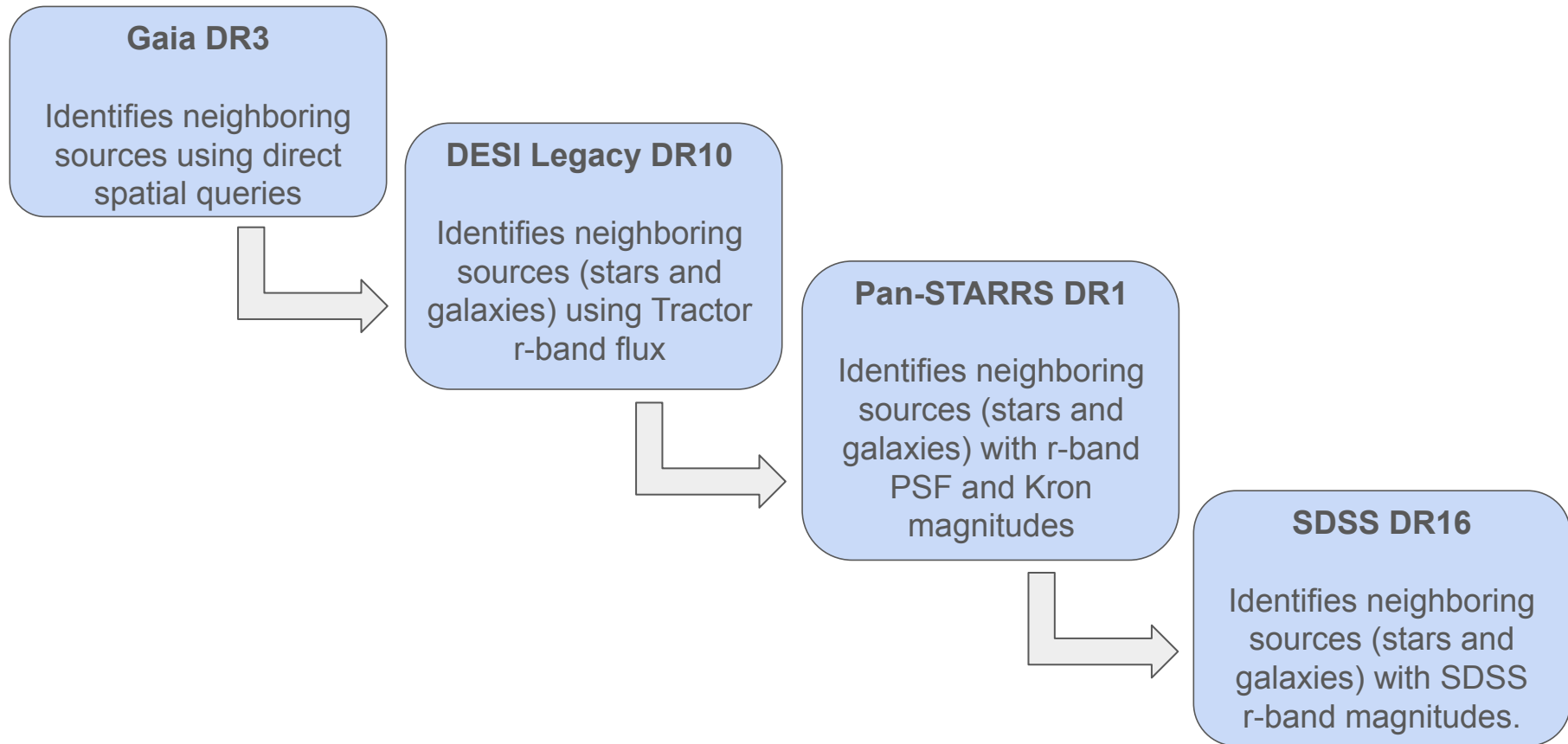


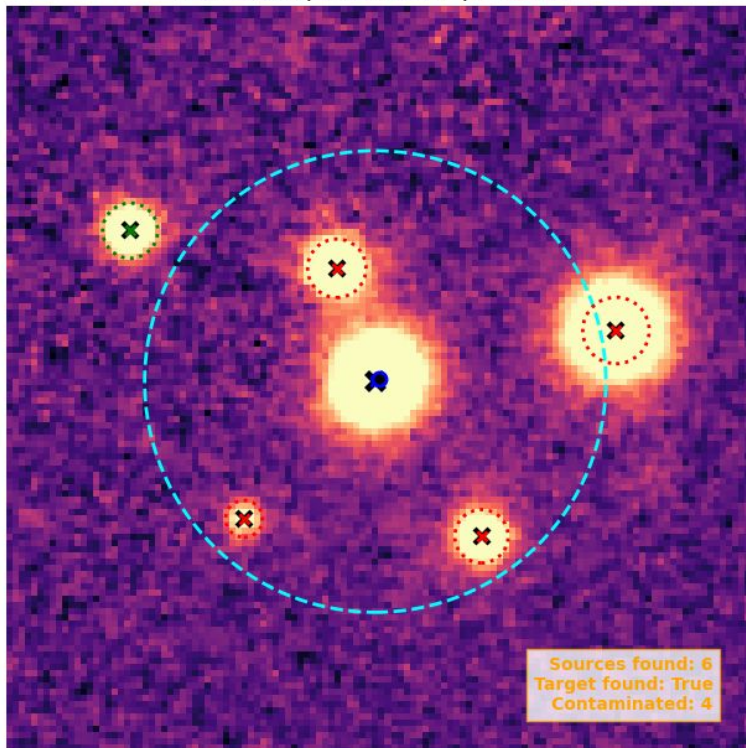
Image Filtering

- For high-confidence cleaning, the code performs direct **image-level checks**:
 - **Magnitude Limit ($G > 13$):**
 - **Faint background contaminants** missed by catalogs do **not contribute** enough flux to significantly bias a bright target.
 - Bright stars frequently **saturate** these optical survey images, which breaks the automated PSF fitting and creates **false positives**
 - **Fallback Hierarchy**: Sequentially fetches the highest-resolution optical FITS cutout available (Legacy ➡ Pan-STARRS ➡ SDSS ➡ DSS2)
 - **PSF Fitting**: Detects all sources within the 30" Field of View (FOV) using DAOSStarFinder
 - **Exclusion Zone**: Evaluates all neighboring detections within the 9.3" radius around the target center.
 - **Dynamic PSF Wings**: Models the physical extent of neighbor light to determine if it bleeds into the target's 9.3" extraction zone.

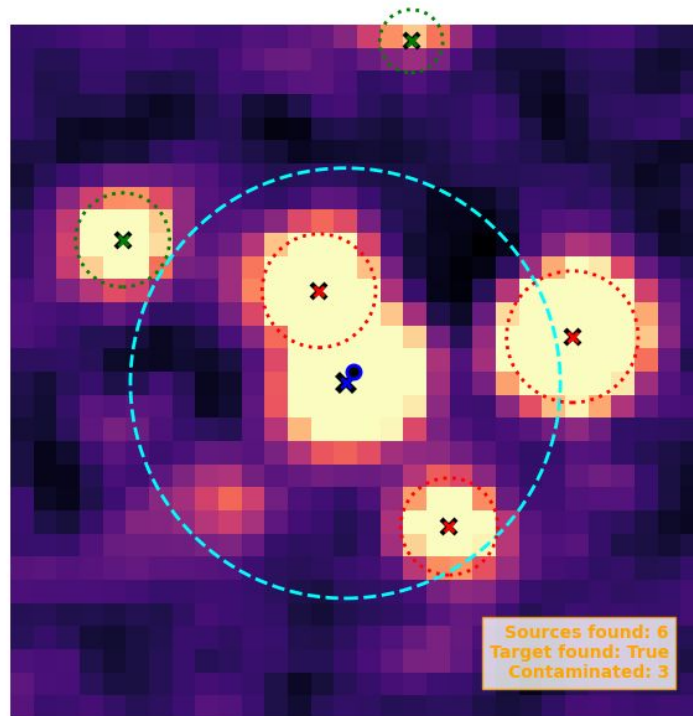
Multi-Survey Image Filtering

Target: 3108936854882971904 | Multi-Survey Astrometric Check

PanSTARRS DR1 (6.1-sigma)
(0.250"/px) | Bounds: 9.3" | Wing: 1.0



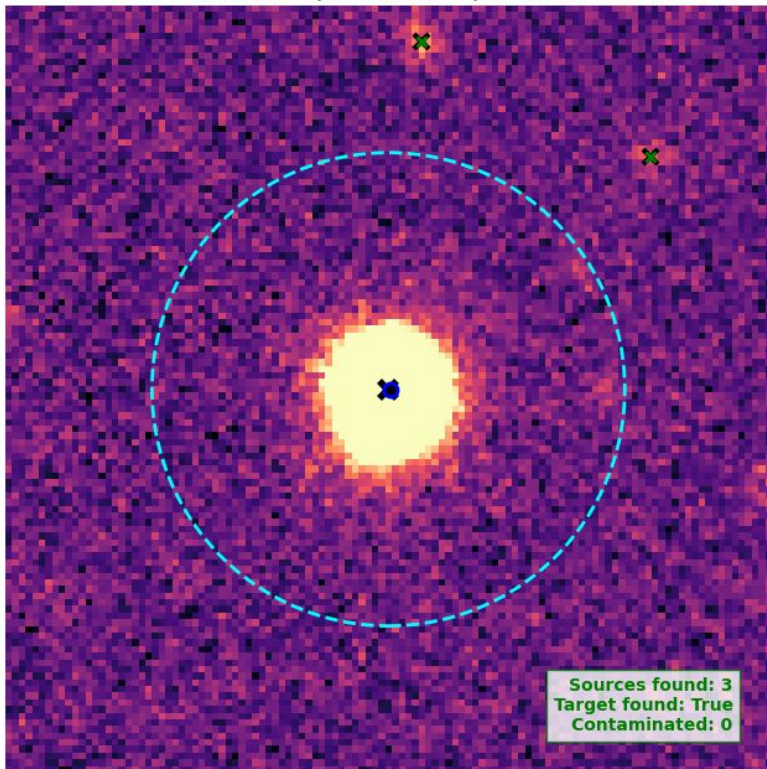
DSS2 (4.0-sigma)
(1.000"/px) | Bounds: 9.3" | Wing: 1.0



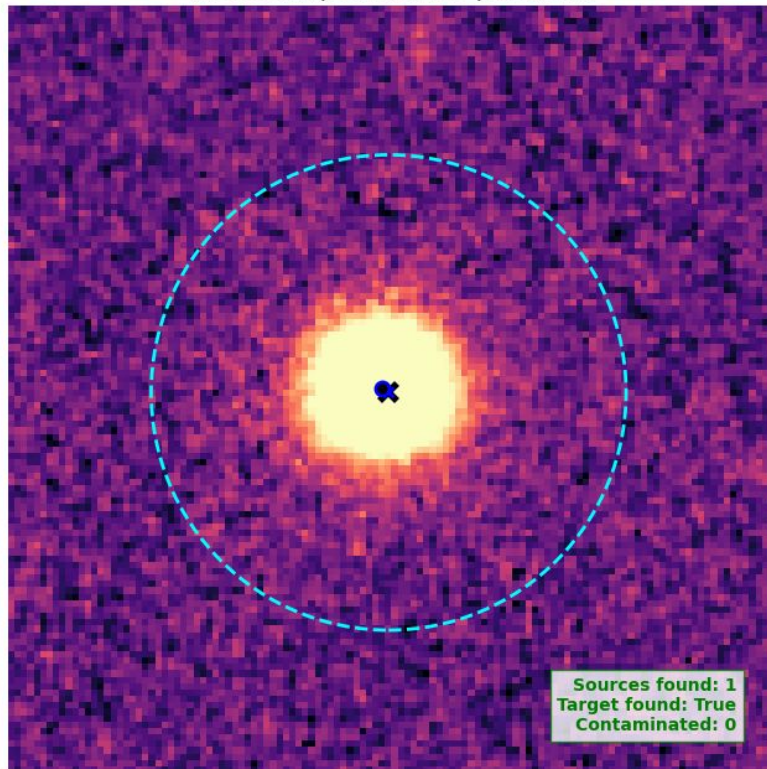
Multi-Survey Image Filtering

Target: 4012093916593061120 | Multi-Survey Astrometric Check

DESI Legacy DR10 (6.1-sigma)
(0.262"/px) | Bounds: 9.3" | Wing: 1.0



PanSTARRS DR1 (6.1-sigma)
(0.250"/px) | Bounds: 9.3" | Wing: 1.0



Spectral Extraction: SPXQuery vs. SPIFF

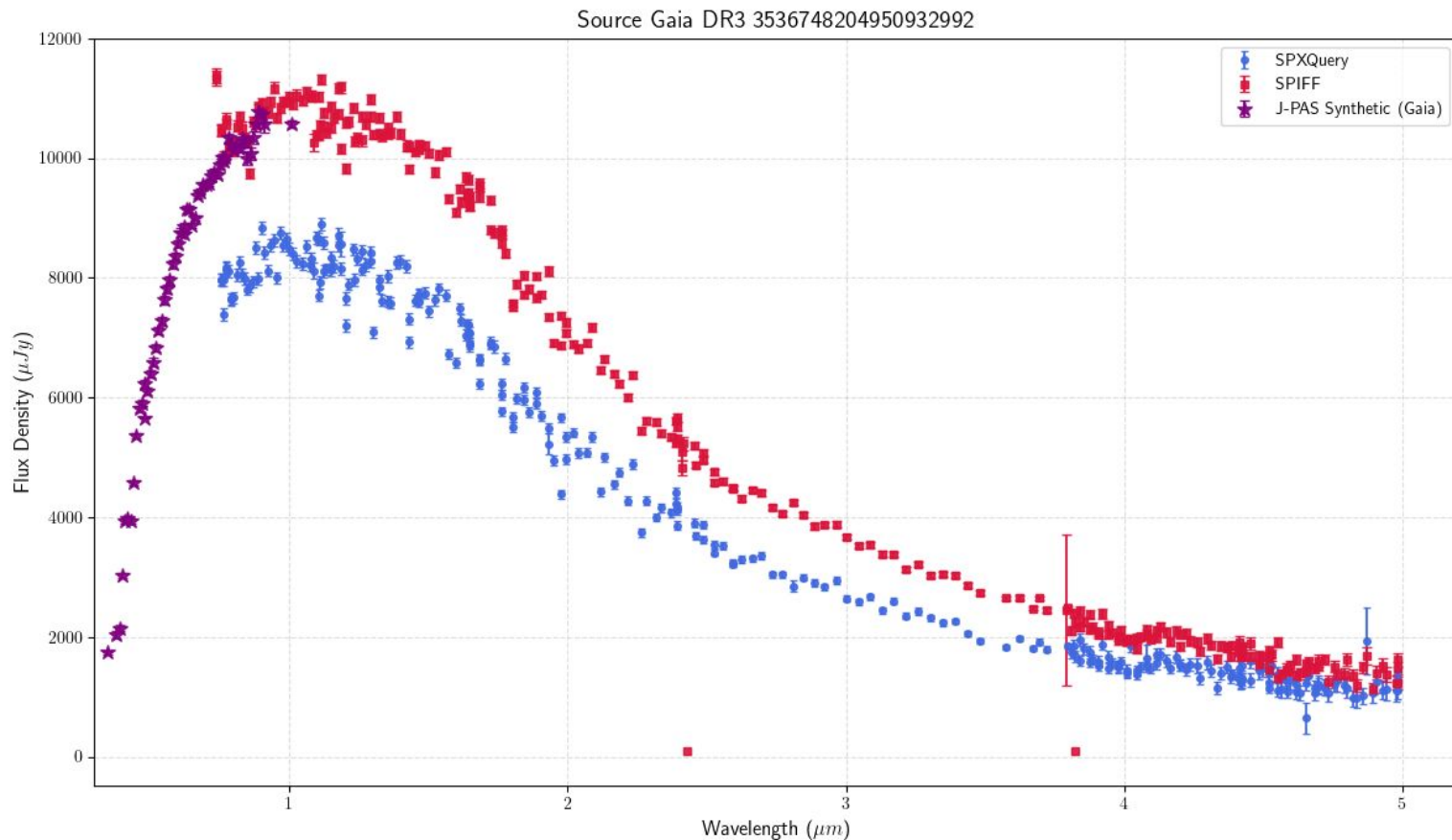
SPXQuery (Fast, Bulk Extraction)

- <https://github.com/WenkeRen/spxquery>
- Automates the full pipeline from querying to photometry using **parallel processing**
- Downloads only localized image cutouts, **reducing storage footprint by ~99%**.
- **The Limitation:** It relies on simple **aperture photometry** (e.g., a **3-pixel / 9.3" radius**). While relatively fast (10 minutes per source), it systematically misses faint light in the PSF wings/tails, underreporting total flux.

SPIFF (Precise, Single-Target Extraction)

- <https://github.com/jgagneastro/SPIFF>
- Uses the exact instrumental **SPHEREx PSF**
- Crucially accounts for target **proper motion** across different SPHEREx epochs
- **The Limitation:** It is **accurate but slow** for large catalogs, taking > 3 hours to extract a single source.

Spectral Extraction: SPXQuery vs. SPIFF



Empirical Calibration Logic

- **Problem:**

- SPXQuery flux is consistently lower than SPIFF flux

- **Solution:**

- We computed **correction factors** for multiple sources across various magnitudes and wavelengths

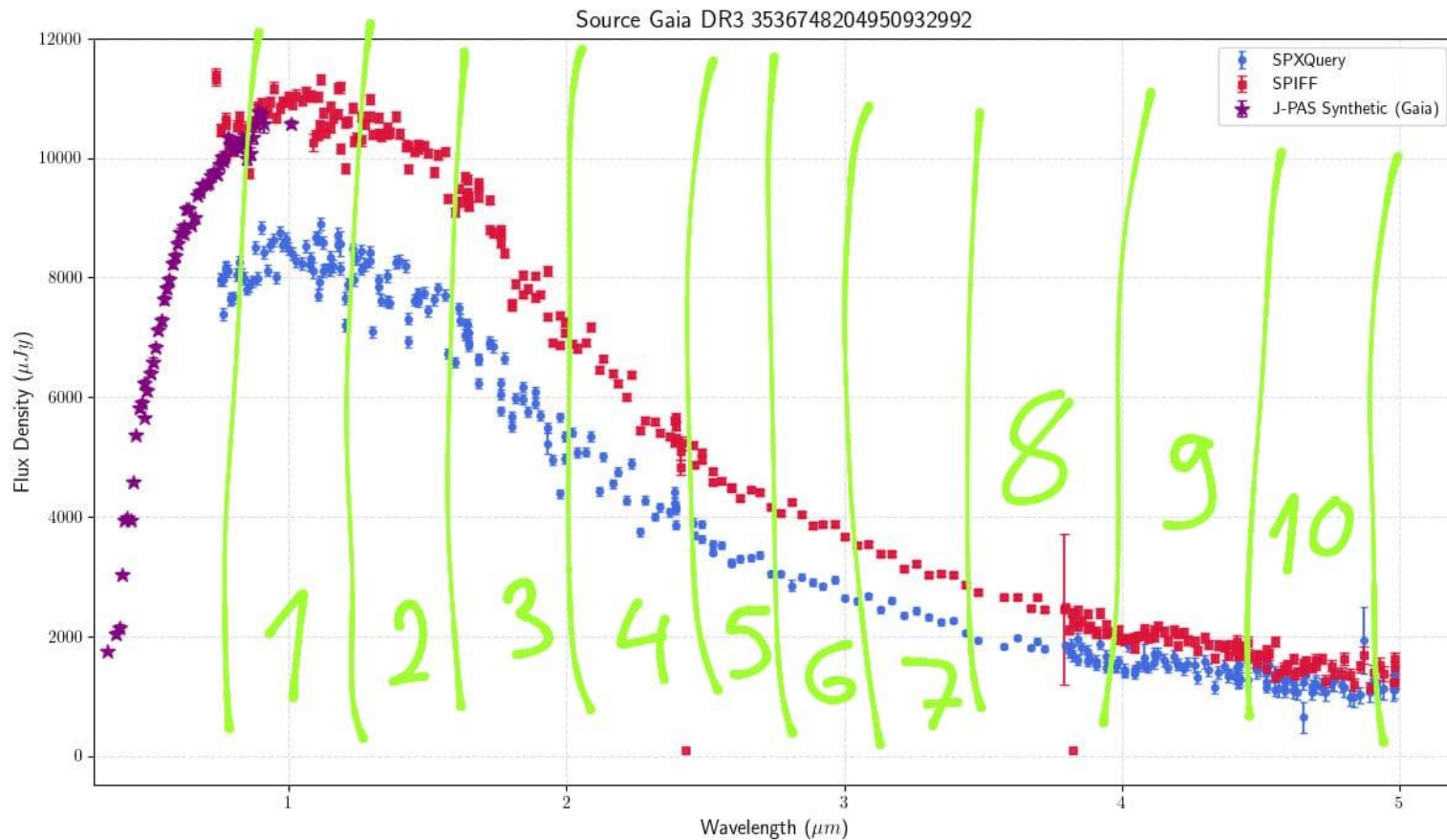
- **Implementation:**

- The pipeline applies a flux-dependent correction across 10 wavelength bins

- **Result:**

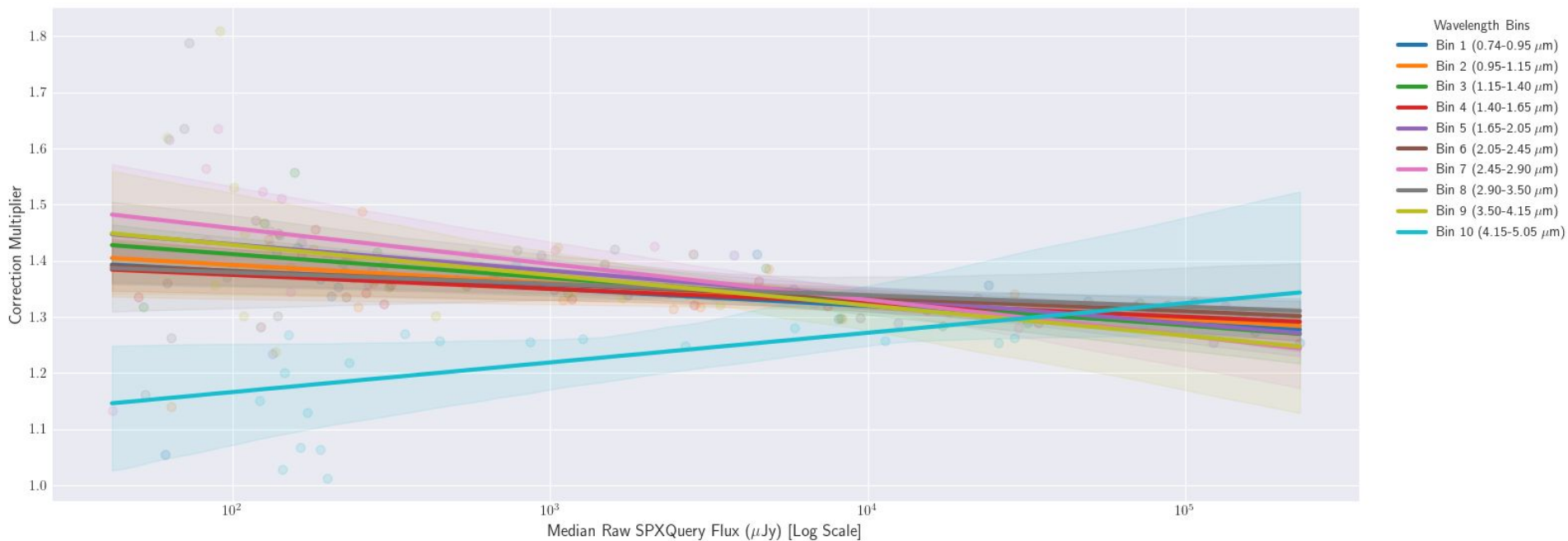
- SPXQuery aperture flux matches SPIFF PSF flux

Empirical Calibration Logic



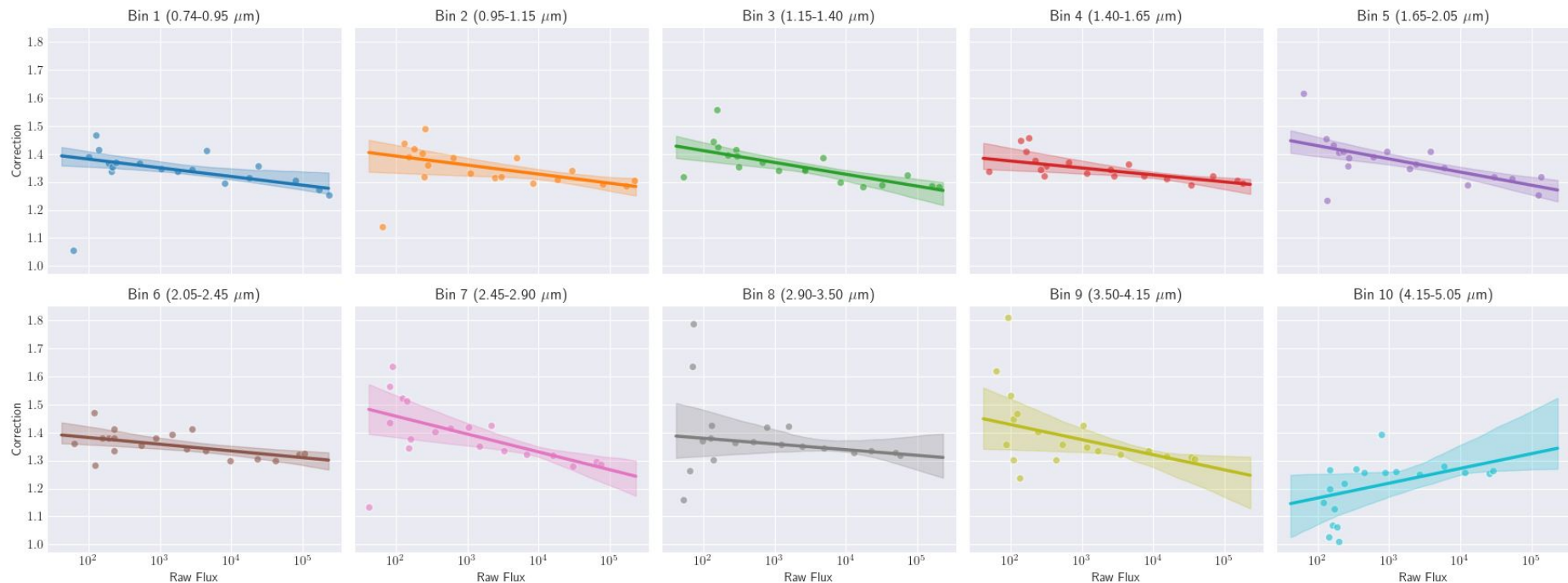
Empirical Calibration Logic

Calibration Curves with 95% Confidence Intervals

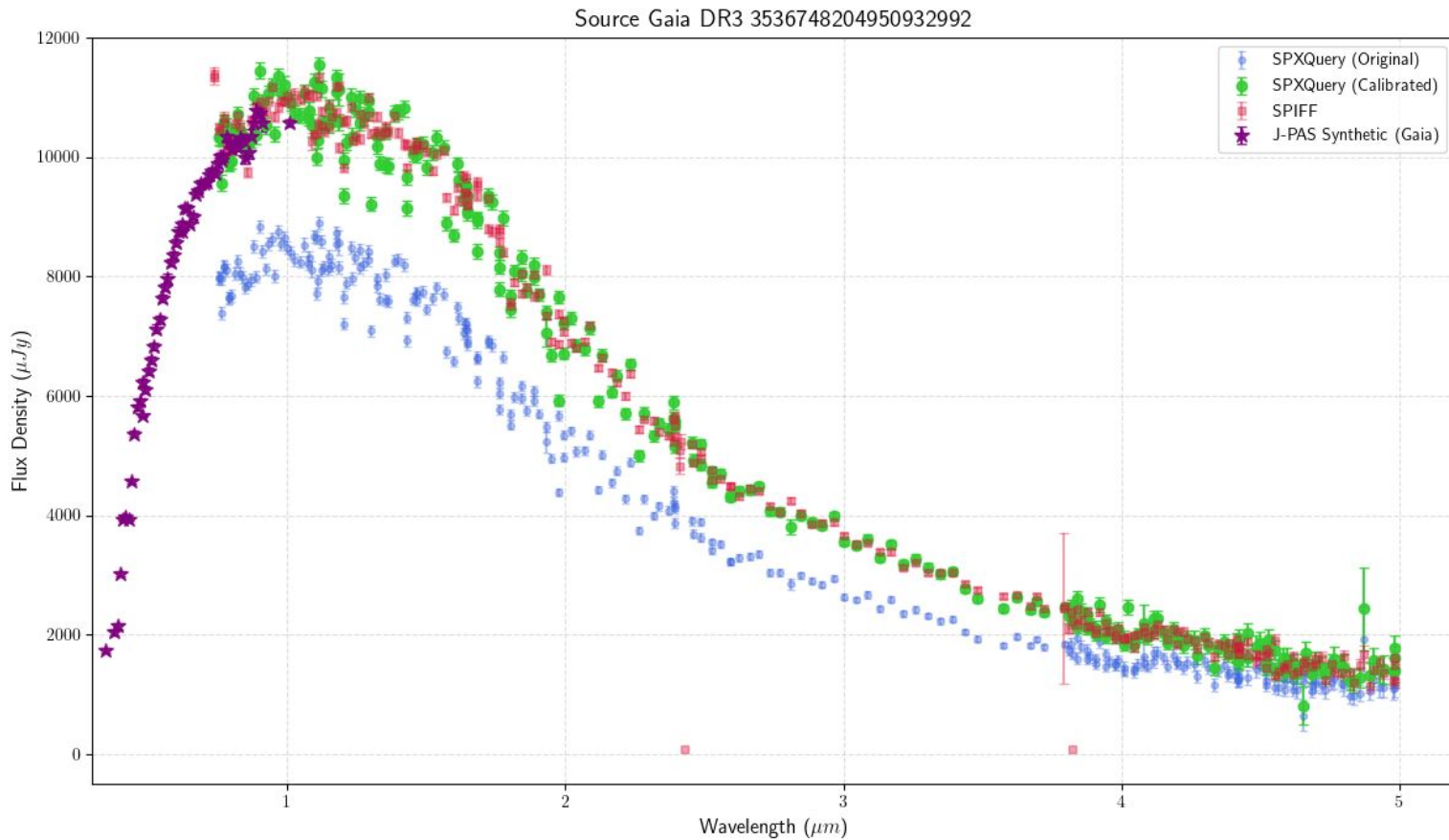


Empirical Calibration Logic

Isolated Calibration Curves with 95% Confidence Intervals per Bin



Calibration Visualized



Summary

- **Clean first, query second**
 - Use the 9.3" radius to match the extraction aperture
- **Code:**
 - https://github.com/andrijazupic/SPHEREx_pipeline